

## EXPERIMENT – 4

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### Demonstration of Hash Cracking Using John the Ripper Tool in Kali Linux

#### Aim

To demonstrate password hash cracking using the John the Ripper tool in Kali Linux for understanding password vulnerabilities and security analysis.

#### Software Requirements

- **Host Operating System:** Windows / Linux / macOS
- **Virtualization Software:** Oracle VirtualBox
- **Guest Operating System:** Kali Linux
- **Security Tool:** John the Ripper
- **System Requirements:**
  - RAM: Minimum 4 GB (8 GB recommended)
  - Disk Space: Minimum 30 GB
  - Processor: Intel/AMD with Virtualization support

#### Introduction

Password hashes are commonly used to store user credentials securely. However, weak passwords and outdated hashing techniques can be exploited using password-cracking tools. John the Ripper is a popular open-source password auditing and cracking tool used to identify weak passwords by performing dictionary, brute-force, and hybrid attacks. Using Kali Linux in a virtualized environment allows safe and ethical experimentation without affecting real systems.

#### Procedure

##### Step 1: Start Kali Linux Virtual Machine

1. Open Oracle VirtualBox.
2. Start the Kali Linux virtual machine.
3. Log in using Kali credentials.

##### Step 2: Open Terminal

1. Click on **Terminal** from the Kali Linux menu.
2. Ensure John the Ripper is installed:
3. `john --version`

##### Step 3: Prepare the Hash File

1. Create a text file containing password hashes (example: hash.txt).
2. Store the hash inside the file:

3. nano hash.txt
4. Save and exit the file.

#### Step 4: Run John the Ripper

1. Execute the John the Ripper command:
2. john hash.txt
3. The tool starts cracking the hash using default wordlists.

#### Step 5: View Cracked Passwords

1. To display cracked passwords:
2. john --show hash.txt

#### Output

John version / installation check:

```
Version: 1.9.0-jumbo-1-bleeding-7e942ec 2019-04-13 14:17:12 +0200
Build: linux-gnu 64-bit x86_64 SSE2 AC OMP
SIMD: SSE2, interleaving: MD4:3 MD5:3 SHA1:1 SHA256:1 SHA512:1
Deploy: sandbox ready
System-wide exec: /snap/john-the-ripper/current/run
System-wide home: /snap/john-the-ripper/current/run
Private home: ~/.john
OMP fallback binary: john-sse2-non-omp
$JOHN is /snap/john-the-ripper/current/run/
Format interface version: 14
Max. number of reported tunable costs: 4
Rec file version: REC4
Charset file version: CHR3
CHARSET_MIN: 1 (0x01)
CHARSET_MAX: 255 (0xff)
CHARSET_LENGTH: 24
SALT_HASH_SIZE: 1048576
SINGLE_IDX_MAX: 32768
SINGLE_BUF_MAX: 4294967295
Effective limit: Max. KPC 32768
Max. Markov mode level: 400
Max. Markov mode password length: 30
gcc version: 5.4.0
GNU libc version: 2.23 (loaded: 2.23)
Crypto library: OpenSSL
OpenSSL library version: 01000207f
OpenSSL 1.0.2g 1 Mar 2016
GMP library version: 6.1.0
Regex library version: 1.4 (loaded: 1.4.0)
File locking: fcntl()
fseek(): fseek
ftell(): ftell
fopen(): fopen
memmem(): System's
```

Hash file:

```
kali:~$ cat lin.txt
kali:$y$j9T$AB7f3W0FqXBj299UsTE0R0$ntPaA0tAgP55AQiBTCSr5R9zxN3E6RF0fBOHhFWXzf5:1000:100
0:Kali,,,:/home/kali:/usr/bin/zsh
user1:$y$j9T$ntKL6I5jJ/LpNG5lmoQiF0$z.mXeK/YlKOr2ix241svrWQJHwsgAzBK.enssb0qyZB:1001:10
01::/home/user1:/bin/sh
user2:$y$j9T$yDFGTveLytT0KtZeJNN7s/$I0eRMLYKDF4KSc2BGC/yAGAJukGoadwEjhXg0sbNkq/:1002:10
02::/home/user2:/bin/sh
user3:$y$j9T$71AzvEcwkG1h3LxCncPxi0$1rcMFwgCSYXrQXl6EEil0zuQv5KT.VtKV/xJCLPaQyB:1003:10
03::/home/user3:/bin/sh

(kali@kali)~$ john --format=crypt --wordlist=rockyou lin.txt
Using default input encoding: UTF-8
Loaded 4 password hashes with 4 different salts (crypt, generic crypt(3) [?/64])
Cost 1 (algorithm [1:descript 2:md5crypt 3:sunmd5 4:bcrypt 5:sha256crypt 6:sha512crypt]
) is 0 for all loaded hashes
Cost 2 (algorithm specific iterations) is 1 for all loaded hashes
Will run 2 OpenMP threads
Press 'q' or Ctrl-C to abort, almost any other key for status
password (user2)
password123 (user1)
Password (user3)
```

John cracking the hash:

```
$ john --incremental --format=raw-sha256 inc.txt
Using default input encoding: UTF-8
Loaded 20 password hashes with no different salts (Raw-SHA256 [SHA256 256/256 AVX2 8x])
Warning: poor OpenMP scalability for this hash type, consider --fork=2
Will run 2 OpenMP threads
Press 'q' or Ctrl-C to abort, almost any other key for status
password
password
admin01
admin05
admin04
admin03
admin02
password1
password2
password3
password7 (daniel1984)
```

john --show hash.txt

## Result

Hash cracking was successfully demonstrated using the John the Ripper tool in Kali Linux. The experiment highlights the importance of using strong passwords and secure hashing algorithms.

## Reference

<https://www.kali.org/tools/john/>

 **Note**

This experiment is conducted **only for educational and ethical purposes** in a controlled laboratory environment.